



Smarter AI-based surveillance cameras boost security and work efficiency (Credit: www.hanwha.com)

size is expected to grow at a CAGR of 10.4 per cent from 45.5 billion dollars in 2020 to 74.6 billion dollars by 2025.

AI taking over passive surveillance solutions

While the installation of CCTV cameras to stream live videos on screens has been done on a massive scale globally, be it in retail stores, hospitals, offices, among others, they are usually passive and only paid attention to after an incident of breach has taken place or there is information about a potential threat. Also, interpretation can be a daunting task as the massive data obtained cannot be utilised to full extent in conventional systems. AI cameras are more intrusive and make real-time monitoring and intelligent analysis of the footage possible that can prevent occurring of such events.

Mostly, low-resolution cameras are utilised for recording, resulting in poor quality images. AI can be used for sharpening unclear images to draw meaningful insights. No matter, the resolution, facial, object, and event recognition become much easier. Recognition can even be done from someone's physical characteristics like height, posture, and build as well as activity patterns.

This is also beneficial since human operators tend to miss out on crucial details, which can have serious consequences, due to reasons such as mental and physical fatigue from continuous monitoring. AI ensures that there are no errors and attention span issues. When remote monitoring in places like factories is required, ideal conditions and performance param-

eters can be programmed into the AI system and it can alert operators, save costs, and prevent machine failures.

Behind the functioning of AI

Machine-based learning algorithms are used to monitor and analyse the images, videos, and data recorded from surveillance solutions. The most common type of AI is rule-based wherein the programmers feed the system with pre-designed rules that must be true in an ideal situation. But slowly the use of behavioural analytics that does not need pre-defined rules is growing. The self-learning system works by auto-detecting and classifying data related to the target environment.

Automated platforms allow smooth collection and storage of information coming in from multiple sources. AI platform Ella by IC Real-time, for instance, makes the data instantly searchable. Users can even narrow down their results by filters like time range and location as well as give their feedback by rating their relevance.

When it comes to surveillance, AI can utilise machine-based vision and be trained to observe every detail in the livestream, or record and notify the operator in case of even the slightest deviation from activity (incidents, behaviour, or actions) that is usual in the ideal scenario that it has been fed, ensuring proactive real-time security. Data fed into the algorithms can include geographic location, biometric data, social media feeds, history of arrest, among others. Deep learning has heavily engineered neural networks

that can detect objects in real-time. Popular methods include faster Region-CNN (RCNN), You Only Look Once (YOLO), and the Single Shot Detector (SSD) networks. The choice of algorithm depends on parameters like speed, accuracy, size, and so on.

For training the surveillance system, training the model is done after data preparation. For the last step—inference, if the processing is done on the edge, onboard GPUs need to be used, or models with lower accuracy need to be used since microcontrollers are not as powerful as GPUs.

Facial recognition uses both video footage and still images for not only a person's identification via database matching but for even more in-depth purposes like expression analysis through crowd scanning and analysis of other similar trends. Evolv Technology, for example, offers a security system that consists of a screening machine and facial recognition application. The screening device detects metallic and non-metallic explosives and other dangerous weapons, while the facial recognition algorithms match the faces on the cameras with the watchlist in the database.

Where can it be used?

When it comes to national security, AI-enabled solutions can detect unattended objects like bags in airports and other places like metros and railway stations, easing the work for security personnel. Also, passengers can be verified through facial recognition along with biometric travel documents. If they do not get cleared, manual screening can be done to meet the requirements. This is even suitable for border control systems.

With mobile surveillance bots like micro drones, large areas of land can be covered. Startup Shield AI, for instance, offers Hivemind Nova drone and machine learning applications that the team says helps military or security personnel in surveillance of even GPS absent high threat environments.

Amongst the main aspect of smart cities is better security by predicting, preventing, addressing, and reducing crimes and other emerging threats to make them safe cities. Smart cities